

Ran Mu

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EDUCATION

Lund University M.S. in Virtual Reality and Augmented Reality	2024.09 – 2026.06
• Coursework: Virtual Reality in Theory and Practice (4/5), Augmented Reality - Interaction (G), Computer Graphics (5/5), High Performance Computer Graphics (4/5), Computer Vision (5/5).	
Beijing Forestry University B.E. in Computer Science and Technology	2017.09 – 2022.06
• Coursework: C++ Programming (97), Algorithm Design & Analysis (89), Operating Systems (90).	

Professional Skills

- **Programming Languages:** Proficient in C++ (OOP, Memory Management, Templates, STL, C++11); experienced in C#.
- **Game Engines:** Full-cycle development in Unity (Gameplay logic, physics, UI, animation) and multiplayer synchronization using Netcode for GameObjects; familiar with Unreal Engine fundamentals.
- **Computer Graphics:** Deep understanding of Rendering Pipelines and Illumination Models; familiar with PBR, Anti-aliasing, and Shadow techniques; mastered Spatial Acceleration Structures (BVH, Quad/Octree); experienced in Shader programming.
- **CS Fundamentals:** Strong grasp of Data Structures and Algorithms; proficient in low-level Memory Allocation and Multi-threading; familiar with core Network Protocols.
- **Architecture & Optimization:** Proficient in Design Patterns (Singleton, Factory, Observer) for games; experienced in CPU/GPU performance tuning and Draw Call batching.
- **Tools:** Blender (Asset processing), Figma (Prototyping), Git (Version control).

PROFESSIONAL EXPERIENCE

Sony <i>Research Intern</i>	2026.01 – Present
• Developed a multi-modal ReID framework for competitive sports by integrating global appearance, jersey numbers, and fine-grained shoe attributes, achieving a Rank-1 accuracy of 92% in complex multi-player tracking scenarios.	
• Implemented a multi-frame temporal aggregation mechanism to mitigate feature degradation caused by severe occlusions, further enhancing the robustness of the identity representation.	
ByteDance <i>Product Intern</i>	2021.07 – 2021.10
• Conducted user flow design for the wrong-answer printing module, translating qualitative insights into technical solutions.	
• Collaborated closely with engineering and design teams to drive rapid prototyping and feature implementation, directly improving end-user review efficiency.	

PROJECT EXPERIENCE

LLM-Driven Mixed Reality Cooking Assistant <i>Unity, Meta XR SDK, Gemini API</i>	2025.09 – 2025.12
• Developed a Meta Quest 3 application that combines Passthrough for environment awareness with Gemini 2.5 Flash to recognize ingredients and generate structured recipes with text and images.	
• Engineered an immersive controller-free system supporting visual, voice, and gesture inputs for intuitive XR interactions.	
Multiplayer Kitchen Simulation Game <i>Unity, Netcode for GameObjects</i>	2024.12 – 2025.03
• Built a multiplayer environment using Hybrid-Authoritative architecture; implemented a full lobby system with Unity Lobby, Authentication, and Relay for stable session management.	
• Leveraged NetworkList and custom RPCs for dynamic state synchronization across distributed clients.	
• Standardized item interactions via Interfaces for scalability, and decoupled data and logic using ScriptableObjects to reduce hard-coding.	
Ray Tracing Renderer <i>C++</i>	2024.10 – 2024.12
• Developed a PBR material system supporting multiple shading models for photorealistic rendering output.	
• Integrated MSAA, Depth of Field, and Gamma Correction to enhance image fidelity and physical accuracy.	
• Implemented BVH acceleration structures to minimize ray-intersection overhead and improve rendering efficiency.	